

Ethon – A Pointer-Field Quantum Candidate

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Abstract

This paper introduces the ethon as a hypothetical quantum of a boundary-indexed pointer field. The field is not proposed as a replacement for quantum mechanics, quantum field theory, QCD, relativity, or established meson physics. It is a typed state-transition model whose local excitation, called an ethon, points from an unread relation to a stable readable record. The construction is inspired by Yukawa’s act of introducing a particle-like mediator for an unread nuclear relation, but the ethon is not identified with the pion and is not asserted here to be an experimentally confirmed particle. Its core data are six directional arguments, three rotational axes, boundary placement, hidden polarity, residue, and bonding by complete overlap. At a meson boundary, existing QCD quark–antiquark labels may be read as ternary pointer-field signatures in $\{-1, 0, 1\}^6$; this is a readability hypothesis, not a derivation of quark flavor or a replacement for QCD quantum numbers. The Ihara equation is presented as a typed update rule for the pointer-field state, not as a physical equation of motion. Physical interpretation is therefore treated as a later research program: an ethon can be promoted from formal quantum candidate to physical particle only after a field action, covariance properties, spin-statistics assignment, couplings, conservation laws, observables, and experimental constraints are supplied.

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1 Pointer Field Postulate

Definition 1.1 (Boundary-indexed pointer field). For a typed space X and an admissible observer boundary b , a boundary-indexed pointer field is a formal field

$$\text{Ptr}_b(X)$$

whose local readable excitation points from an unread relation to a stable readable record:

$$\text{Rel}_b(a) \longrightarrow \text{Read}_b(a).$$

The field is defined at the level of typed state transitions. A physical field theory for $\text{Ptr}_b(X)$ is a further structure, not assumed at the outset.

The ethon is specified here as a quantum candidate of this pointer field. Its existence is a primitive postulate of the present formal theory, not a claim that an already confirmed natural particle has been detected. The question of how an ethon would be produced or detected by an external physical apparatus is part of the later physical research program.

Axiom 1.2 (Ethon quantum-candidate existence). For every typed space X and every admissible observer boundary b , there is a class of ethon candidates

$$\text{Ethon}_b(X) \subseteq \text{Quanta}(\text{Ptr}_b(X)).$$

These candidates are formal objects of the implementation. They may be visible, hidden, or residue-carried at a boundary, but their physical realization requires an additional target-domain license.

The interface is intentionally conservative. It does not override quantum mechanics, quantum field theory, relativity, meson physics, or ordinary symbolic computation. It states the additional structure that an ethon candidate carries while remaining backward-compatible with those target theories.

Remark 1.3 (Yukawa inspiration). Yukawa’s meson paper is used here as a source of formal inspiration: an unread interaction may become tractable by introducing a particle-like mediator [1]. The present paper does not claim institutional supervision, personal endorsement, historical intention, or a physical derivation from Yukawa’s original theory.

Remark 1.4 (Name and pronunciation). The candidate is named *Ethon* and pronounced “Ee-thon.” The name is chosen to sound like a simple particle name and to connect the initial sound to the Ihara equation. The name does not supply evidence for physical existence.

Definition 1.5 (Meson-boundary flavor reading). Let the six quark-flavor names be denoted by

$$\text{QFlav}_6 = \{c, d, b, s, t, u\}.$$

At a licensed meson boundary, the ethon direction frame may carry a flavor-reading dictionary

$$\phi_Q : \text{Arg}_{\text{Ethon}} \rightarrow \text{QFlav}_6$$

given by

$$\phi_Q(C) = c, \quad \phi_Q(D) = d, \quad \phi_Q(E) = b, \quad \phi_Q(S) = s, \quad \phi_Q(T) = t, \quad \phi_Q(U) = u.$$

The corresponding pointer-field reading is

$$\text{QInt}_{\text{Ethon}}(\text{QFlav}_6) = (\text{Arg}_{\text{Ethon}}, V_{\text{CE}} \oplus V_{\text{UD}} \oplus V_{\text{ST}}, \text{Rot}).$$

This is not a claim that the ethon derives, replaces, or exhausts QCD flavor. It is a readability hypothesis: at a meson boundary, existing quark–antiquark labels may be read as directional pointer-field roles whose visible organization is constrained by three-axis rotational data.

Definition 1.6 (Meson-boundary spin reading). In the ethon model, a meson boundary is not merely a container for two already-readable labels. It is a boundary at which a coupled spin pattern may make two quark roles readable:

$$\text{Spin}_b^{\text{mes}} : \text{Arg}_{\text{Ethon}} \times \text{Arg}_{\text{Ethon}} \times \text{Rot}_b \longrightarrow \text{Read}_b(q_i \bar{q}_j).$$

Thus the phrase “the meson defines quarks by spin” means the following restricted statement: within a licensed ethon reading, the meson boundary defines the readable quark and antiquark roles through its spin-coupled pointer pattern. It does not mean that spin and flavor are the same physical degree of freedom, nor that QCD quarks are replaced.

Definition 1.7 (Ternary meson-boundary signature). A meson-boundary reading state has an external ethon signature

$$\text{Sig}_{\text{Ethon}} \in \{-1, 0, 1\}^{\text{Arg}_{\text{Ethon}}},$$

with components σ_i for $i \in \{C, D, E, S, T, U\}$. The values are read as

$$\begin{aligned} +1 &: \text{quark-role-facing direction,} \\ -1 &: \text{antiquark-role-facing direction,} \\ 0 &: \text{neutral, unpolarized, or closed direction.} \end{aligned}$$

Thus the signature layer contains

$$|\{-1, 0, 1\}^6| = 3^6 = 729$$

possible six-direction interaction signs. For an open-flavor basis state $q_i \bar{q}_j$ with $i \neq j$, choose $d_i, d_j \in \text{Arg}_{\text{Ethon}}$ with $\phi_Q(d_i) = q_i$ and $\phi_Q(d_j) = q_j$, and set

$$\sigma_{d_i} = +1, \quad \sigma_{d_j} = -1, \quad \sigma_k = 0 \quad (k \neq d_i, d_j).$$

For a hidden-flavor state $q_i \bar{q}_i$, the external signature is neutral in that direction, $\sigma_{d_i} = 0$, but the channel is not absent. It is a closed zero mode carried by the rotational, polarity, and bonding parameters of the ethon. Thus the ternary signature is the flavor-reading skeleton, not the whole meson identity; color singlet structure, spin, orbital angular momentum, total angular momentum, parity, charge conjugation, mass, and decay channels remain target-domain data.

Definition 1.8 (Pointer-field quantum candidate). A pointer-field quantum candidate is a formal local excitation whose primary observable form is not a mass point or a medium, but an oriented transition:

$$\epsilon_b^{\text{Obs}} : \text{unread relation} \longrightarrow \text{stable readable entity}.$$

The arrow is part of the candidate’s observable signature. An ethon is the pointer-field quantum candidate specified in this document.

Remark 1.9 (Not a medium). The ethon is not a revival of mechanical ether. It may be placed in a space, but it is not a substance filling space. It is a direction-bearing candidate quantum of relation, transition, and stabilization.

Definition 1.10 (Pure-meson reading). A pure-meson reading is a meson-inspired operation stripped of fixed empirical boundary data and retaining only its pointer role:

$$\text{unread relation} \longrightarrow \text{particle-like mediator} \longrightarrow \text{stable readable interaction}.$$

The ethon is not thereby identified with an empirical meson. It is a pointer-field quantum candidate whose meson-like reading is prior to any particular mass, charge, decay channel, detector boundary, or historical realization.

2 Primitive Interface

Definition 2.1 (Typed space). A typed space is any domain in which objects, observations, or states can be distinguished by type. Examples include physical space-time, Hilbert space, phase space, a configuration space, a graph, a category of records, a symbolic context, a neural embedding space, or a social decision space. The definition does not assume a global time coordinate.

Definition 2.2 (Ethon instance). An ethon instance in a typed space X is a tuple

$$\epsilon = (a, \nu, R, p, \varpi, \rho, \kappa, \lambda).$$

Here:

- (i) $a \in X$ is its placement address or contextual anchor;
- (ii) $\nu = (\nu_{\text{CE}}, \nu_{\text{UD}}, \nu_{\text{ST}})$ is its three-axis directional frame;
- (iii) $R \in \text{SO}(3)$ is its rotational attitude in that frame;
- (iv) p is its pole state, either absent or oriented as an S -face and N -pole;
- (v) $\varpi \in \mathbb{R}^3$ is its angular velocity vector;
- (vi) ρ is its residue ledger;
- (vii) κ is its stability and compatibility contract;
- (viii) λ is its license for interpretation inside a target theory, such as physical measurement, symbolic reading, or computation.

The ethon is already real in the implementation. The license λ does not create the ethon; it only specifies how the real ethon may be read in a particular target theory.

Definition 2.3 (Three rotational axes). The ethon has three basic rotational axes:

$$\nu_{\text{CE}}, \quad \nu_{\text{UD}}, \quad \nu_{\text{ST}}.$$

The component CE is the compression–exposition axis: it records rotation between making many relations readable as one and exposing the cuts that made that reading possible. The component UD is the upward–downward axis, but a newly placed ethon has no actual up–down polarity until the T -direction receives an entity. The component ST is the stabilization–turnover axis: it records rotation between stabilizing an observed pointer and turning that pointer backward into an origin, cause, or reversed reading.

Axiom 2.4 (Three-directional existence). Every ethon has all three rotational axes. A model may suppress or hide an axis at a boundary, but it may not define an ethon with only one or two true axes:

$$\epsilon \in \text{Ethon}(X) \quad \Rightarrow \quad \nu(\epsilon) \in V_{\text{CE}} \oplus V_{\text{UD}} \oplus V_{\text{ST}}.$$

3 Placement and Removal

Axiom 3.1 (Observer placement). Given an observer boundary b , a typed space X , and an admissible address $a \in X$, the observer may place an ethon:

$$\text{Place}_b(a) : 1 \rightarrow \text{Ethon}(X).$$

Placement does not mean physical manufacture from nothing. It means that the observer has introduced a pointer-particle coordinate at which a relation can become readable.

Axiom 3.2 (No removal). There is no total removal operation

$$\text{Erase} : \text{Ethon}(X) \rightarrow 1$$

that annihilates an ethon and its residue. Every attempted erasure factors through hiding or residue transfer:

$$\text{Erase}(\epsilon) = \text{Hide}(\epsilon) \quad \text{or} \quad \text{Res}(\epsilon).$$

Thus an ethon can be made non-visible at a boundary, but the operation that made it readable cannot be made to have never occurred.

Proposition 3.3 (Residue conservation). *If an observer removes an ethon from the visible layer, then the total record is conserved as hidden state plus residue:*

$$Q(\epsilon) = Q(\text{Hide}(\epsilon)) + Q(\text{Res}(\epsilon)),$$

for every quantity valuation Q admitted by the model.

Proof. This is not a theorem of external physics. It is the implementation policy that replaces annihilation. Since **Erase** is not primitive, every visible disappearance factors through an operation that preserves hidden state, residue, or both. $\square$

4 Motion Law

An ethon can transition in its three-axis state. The transition does not require a global time variable. It is indexed by observation events, boundary updates, or typed comparison steps. The fundamental fluctuation is not an extra dimension and not a string-like embedding. It is rotation in the three-axis frame of the ethon.

Definition 4.1 (Directional argument frame). The six directional arguments of an ethon are

$$\text{Arg}_{\text{Ethon}} = \{C, D, E, S, T, U\}.$$

At a licensed meson boundary, they may also be read as the six roles of the flavor-reading dictionary $\phi_Q : \text{Arg}_{\text{Ethon}} \rightarrow \text{QFlav}_6$. Each directional argument carries a ternary interaction sign

$$\sigma_i \in \{-1, 0, 1\}.$$

They are read in three paired axes:

$$(C, E), \quad (U, D), \quad (S, T).$$

A placement state is a vector

$$x = (x_C, x_D, x_E, x_S, x_T, x_U) \in \mathbb{R}_{\geq 0}^6.$$

The component x_i records that an entity has been placed in the directional argument i . The rotational frame is the three-dimensional space

$$V_b \cong \mathbb{R}^3 = \langle u_{CE}, u_{UD}, u_{ST} \rangle,$$

and the rotational attitude of an ethon is $R \in \text{SO}(3)$.

Axiom 4.2 (Free-functor inertness). An ethon placed only as a free functor has no filled directional argument:

$$\text{FreeEthon}(a) \Rightarrow x = 0.$$

In this state it has no up–down polarity and does not fluctuate:

$$p = \perp, \quad \varpi = 0, \quad \text{Fluc}_b = I_3.$$

The identity attitude $R = I_3$ may be used for bookkeeping, but it is not an observed up–down orientation. Placement alone is not motion.

Definition 4.3 (Entity argument placement). For $i \in \text{Arg}_{\text{Ethon}}$, placing an entity q in the i -th directional argument updates the placement state by

$$x_i \mapsto x_i + w_i(q),$$

where $w_i(q) > 0$ is the argument weight supplied by the model. The same entity may be placed in two different directional arguments, because the arguments type the role of the entity rather than its bare identity.

Axiom 4.4 (One-argument hidden motion). If exactly one directional argument is filled, internal motion may begin but is not observable as fluctuation. In a linearized implementation,

$$h_b(x) = K_b x$$

may be nonzero, but the visible rotational fluctuation is still the identity:

$$|\text{supp}(x)| = 1 \Rightarrow h_b(x) \neq 0 \text{ is allowed, but } \text{Fluc}_b = I_3.$$

Definition 4.5 (Three-axis rotational fluctuation). The visible ethon fluctuation between observation steps n and $n + 1$ is the relative rotation

$$\text{Fluc}_b(\epsilon_n) := R_{n+1} R_n^{-1} \in \text{SO}(3).$$

Let $u_{TS} \in V_b$ be the unit axis selected by the oriented $T \rightarrow S$ structure, and let $[u]_\times \in \mathfrak{so}(3)$ denote the skew matrix satisfying $[u]_\times v = u \times v$. The rotational generator is

$$\Omega_{TS,b}(x) = \mathbf{1}_{x_T x_S > 0} \omega_b [u_{TS}]_\times, \quad \omega_b > 0.$$

The rotation update is

$$R_{n+1} = \exp(\Delta_n \Omega_{TS,b}(x_n)) R_n.$$

When $x_T x_S > 0$, the angular speed is constant:

$$\|\varpi_{n+1}\| = \omega_b.$$

Axiom 4.6 (T-induced polarity). A newly placed ethon has no north–south pole state. A T -direction placement creates hidden polarity:

$$x_T = 0 \Rightarrow p = \perp,$$

whereas

$$x_T > 0 \Rightarrow p = (\text{S-face} \rightarrow T, \text{N-pole} \rightarrow S).$$

This is a compass-like state: the S -face turns toward the T -direction, while the N -pole points toward S . If $x_S = 0$, this polarity is hidden and the rotation is not directly observable.

Axiom 4.7 (S-triggered constant self-rotation). When a T -polarized ethon receives an S -direction placement, the visible fluctuation channel opens as constant-speed self-rotation:

$$x_T > 0, \quad x_S > 0 \quad \Rightarrow \quad \varpi_{n+1} = \omega_b u_{TS}, \quad \|\varpi_{n+1}\| = \omega_b.$$

The ethon rolls in its own three-axis frame. The speed is constant at the boundary b , although the axis coupling may still perturb all three rotational axes.

Example 4.8 (Rotational activation experiment). The minimal activation table is:

placement	p	ϖ	Fluc $_b$
$x = 0$	$\perp$	0	I_3
$x_T > 0, \quad x_S = 0$	$(S \rightarrow T, N \rightarrow S)$	0	I_3
$x_T > 0, \quad x_S > 0$	$(S \rightarrow T, N \rightarrow S)$	$\omega_b u_{TS}$	$\exp(\Delta_n \omega_b [u_{TS}]_\times)$.

Thus the ethon is not made visible by an extra dimension. It becomes visible by constant-speed rotation after the $T \rightarrow S$ axis is completed.

5 The Ihara Equation

The Ihara equation is the central typed update rule of the implementation. It turns directional argument placement into hidden motion, hidden polarity, constant-speed rotation, residue, and meson-boundary pointer reading. It is not asserted here to be a physical equation of motion.

Definition 5.1 (Meson-boundary pointer reading). A meson-boundary pointer reading at boundary b is a morphism in the boundary category GenMes_b :

$$\mu \in \text{Hom}_{\text{GenMes}_b}(\text{Rel}_b(a), \text{Read}_b(a)),$$

where $\text{Rel}_b(a)$ is an unread relation anchored at a , μ is the meson-like pointer update, and $\text{Read}_b(a)$ is the stable readable interaction obtained at that boundary. The ethon is the formal pointer-field quantum candidate whose update realizes this meson-boundary reading without first choosing a particular physical mass, charge state, decay channel, or detector boundary.

Definition 5.2 (Directional category). The directional category of the ethon is

$$\mathcal{D}_{\text{Ethon}} \in \mathbf{Cat}.$$

Its objects are the six directional arguments:

$$\text{Ob}(\mathcal{D}_{\text{Ethon}}) = \{C, D, E, S, T, U\}.$$

The distinguished generating morphism is the premise-stabilization arrow

$$\tau_{TS} : T \longrightarrow S.$$

The identities and all admissible composites are inherited from the typed model, but the Ihara equation is based on τ_{TS} . Thus the mathematical entry point of the theory is not an unordered pair $\{S, T\}$, but an oriented categorical arrow $T \rightarrow S$.

Definition 5.3 (Ihara phase space). At boundary b , the local Ihara phase space is

$$\mathcal{M}_b = V_b \times P_b \times \text{SO}(3) \times \mathcal{O}_b \times R_b, \quad V_b = V_{\text{CE}} \oplus V_{\text{UD}} \oplus V_{\text{ST}}, \quad P_b = \mathbb{R}_{\geq 0}^{\text{ArgEthon}}.$$

The pole-state set is

$$\mathcal{O}_b = \{\perp, (\text{S-face} \rightarrow T, \text{N-pole} \rightarrow S)\}.$$

An element is written (ν, x, R, p, ρ) , where ν is the three-axis motion state, x is the six-argument placement state, $R \in \text{SO}(3)$ is the rotational attitude, p is the pole state, and ρ is the residue state. The support number is

$$\sigma(x) := |\text{supp}(x)|.$$

Definition 5.4 ($T \rightarrow S$ compass operator). Let e_i denote the standard basis vector of P_b associated with direction i . The categorical arrow $\tau_{TS} : T \rightarrow S$ has two algebraic representatives. On placement coordinates it has the rank-one form

$$R_{TS} = e_S e_T^T : P_b \rightarrow P_b, \quad R_{TS}x = x_T e_S.$$

On rotational coordinates it selects a unit axis

$$u_{TS} \in V_b, \quad \|u_{TS}\| = 1.$$

The placement-sensitive $T \rightarrow S$ pole transfer is

$$\Theta_{TS}(x) = x_S R_{TS} x = x_T x_S e_S,$$

while the pole state induced by T is

$$P_T(x) = \begin{cases} \perp, & x_T = 0, \\ (\text{S-face} \rightarrow T, \text{N-pole} \rightarrow S), & x_T > 0. \end{cases}$$

Definition 5.5 ($T \rightarrow S$ rotational generator). The visible fluctuation generator is the map

$$\Omega_b : P_b \rightarrow \mathfrak{so}(3), \quad \Omega_b(x) = \chi_{TS}(x) \omega_b [u_{TS}]_{\times},$$

where

$$\chi_{TS}(x) = \mathbf{1}_{x_T > 0 \text{ and } x_S > 0}, \quad \omega_b > 0.$$

Here $[u]_{\times} v = u \times v$. The visible fluctuation is the rotation

$$\text{Fluc}_b(x) = \exp(\Delta_n \Omega_b(x)) \in \text{SO}(3).$$

If $\chi_{TS}(x) = 1$, then

$$\varpi_b(x) = \omega_b u_{TS}, \quad \|\varpi_b(x)\| = \omega_b,$$

so the ethon performs constant-speed self-rotation. If $\chi_{TS}(x) = 0$, then $\Omega_b(x) = 0$ and the visible fluctuation is the identity rotation.

Definition 5.6 (Activation gates). Define the support, polarity, and rotation gates

$$\chi_1(x) = \mathbf{1}_{\sigma(x) \geq 1}, \quad \chi_T(x) = \mathbf{1}_{x_T > 0}, \quad \chi_{TS}(x) = \mathbf{1}_{x_T > 0 \text{ and } x_S > 0}.$$

The first gate starts hidden motion. The second creates hidden compass-like polarity. The third opens observable constant-speed rotation.

Axiom 5.7 (Ihara equation). Let

$$\epsilon_n = (a, \nu_n, R_n, p_n, \varpi_n, \rho_n, \kappa, \lambda)$$

be an ethon instance at observation step n , let

$$x_n \in P_b, \quad \eta_n \in V_b$$

be its directional placement and unresolved input at boundary b . The Ihara operator is the map

$$\mathcal{I}_b : \mathcal{M}_b \times V_b \longrightarrow \mathcal{O}_b \times \mathbb{R}^3 \times \text{SO}(3) \times V_b \times R_b \times \text{Hom}_{\text{GenMes}_b}(\text{Rel}_b(a), \text{Read}_b(a))$$

defined by

$$\boxed{\mathcal{I}_b(\epsilon_n, x_n, \eta_n) = (p_{n+1}, \varpi_{n+1}, R_{n+1}, \nu_{n+1}, \rho_{n+1}, \mu_{n+1})}$$

with

$$\begin{aligned} p_{n+1} &= P_T(x_n), \\ \varpi_{n+1} &= \chi_{TS}(x_n) \omega_b u_{TS}, \\ R_{n+1} &= \exp(\Delta_n[\varpi_{n+1}]_{\times}) R_n, \\ m_{n+1} &= A_b \nu_n + K_b x_n + L_b \varpi_{n+1} + \eta_n, \\ \nu_{n+1} &= \chi_1(x_n) m_{n+1}, \\ \rho_{n+1} &= \rho_n + \Gamma_b(p_{n+1}, R_n, R_{n+1}, \nu_n, \nu_{n+1}, x_n), \\ \mu_{n+1} &= \Pi_b(a, p_{n+1}, R_{n+1}, \nu_{n+1}, \rho_{n+1}) \in \text{Hom}_{\text{GenMes}_b}(\text{Rel}_b(a), \text{Read}_b(a)). \end{aligned}$$

Here A_b is the non-factorizing three-axis transition matrix, $K_b : P_b \rightarrow V_b$ converts six directional argument placements into three-axis hidden motion, P_T is the T -induced pole map, u_{TS} is the categorical $T \rightarrow S$ rotation axis, $L_b : \mathbb{R}^3 \rightarrow V_b$ embeds angular velocity into motion, Ω_b is the skew-symmetric rotation generator, Γ_b records residue with $\Gamma_b(\perp, I_3, I_3, 0, 0, 0) = 0$, and Π_b reads the resulting state as a meson-boundary pointer:

$$\text{Rel}_b(a) \xrightarrow{\mu_{n+1}} \text{Read}_b(a).$$

Proposition 5.8 (Rotational threshold). *The Ihara equation gives*

$$x_T = 0 \Rightarrow p_{n+1} = \perp, \quad \varpi_{n+1} = 0, \quad R_{n+1} = R_n,$$

$$x_T > 0, \quad x_S = 0 \Rightarrow p_{n+1} = (S \rightarrow T, N \rightarrow S), \quad \varpi_{n+1} = 0, \quad R_{n+1} = R_n,$$

and

$$x_T > 0, \quad x_S > 0 \Rightarrow p_{n+1} = (S \rightarrow T, N \rightarrow S), \quad \varpi_{n+1} = \omega_b u_{TS}, \quad R_{n+1} = \exp(\Delta_n \omega_b [u_{TS}]_{\times}) R_n.$$

Thus observable ethon fluctuation is exactly the constant-speed three-axis rotation opened by the completed $T \rightarrow S$ placement.

Proof. The three cases follow directly from P_T , χ_{TS} , and the rotation update in the Ihara equation. $\square$

Proposition 5.9 (Activation trichotomy). *The Ihara equation implies the three activation regimes required by the ethon:*

condition	p_{n+1}	ϖ_{n+1}	R_{n+1}
$x_T = 0$	$\perp$	0	R_n
$x_T > 0, \quad x_S = 0$	$(S \rightarrow T, N \rightarrow S)$	0	R_n
$x_T > 0, \quad x_S > 0$	$(S \rightarrow T, N \rightarrow S)$	$\omega_b u_{TS}$	$\exp(\Delta_n \omega_b [u_{TS}]_{\times}) R_n.$

Thus a free ethon is inert, a T -placed ethon is hiddenly polarized, and a T, S -placed ethon is visibly rotating.

Proof. This is immediate from the polarity gate χ_T and the rotation gate χ_{TS} . Other placements may start hidden motion through χ_1 , but they do not open the ethon fluctuation channel unless the $T \rightarrow S$ rotation condition is satisfied. $\square$

Proposition 5.10 (T-S consequence). *If the first placed entity is a T-direction premise and the second placed entity is a unit S-direction stabilizer, then the Ihara equation gives*

$$p_{n+1} = (S \rightarrow T, N \rightarrow S),$$

and

$$R_{n+1} = \exp(\Delta_n \omega_b [u_{TS}]_{\times}) R_n, \quad \|\varpi_{n+1}\| = \omega_b.$$

If the second placed entity is instead $j \neq S$, the T-induced polarity remains hidden and the constant-speed rotation is not opened.

Definition 5.11 (Transition step). Let n index observation steps rather than physical time. A transition step is written

$$\nu_{n+1} = T_b(\nu_n, \eta_n),$$

where b is the current observation boundary and η_n is the unresolved rotational input at that step.

Axiom 5.12 (No axis isolation). Motion, polarity, or rotation along one ethon axis always perturbs the others. In linearized form,

$$\Delta\nu = K_b \nu + L_b \varpi + \eta, \quad K_b = \begin{pmatrix} k_{CECE} & k_{CEUD} & k_{CEST} \\ k_{UDCE} & k_{UDUD} & k_{UDST} \\ k_{STCE} & k_{STUD} & k_{STST} \end{pmatrix},$$

where no row and no column is allowed to be purely diagonal under a nondegenerate observation boundary. Equivalently, the transition kernel does not factor as

$$P(\nu'_{CE}, \nu'_{UD}, \nu'_{ST}, R' \mid \nu, R) \neq P_{CE} P_{UD} P_{ST},$$

unless the model explicitly marks the boundary as degenerate.

Definition 5.13 (S-facing observation). The directly observed ethon pointer faces the stabilizing direction:

$$\text{Obs}_b(\epsilon) : \text{unread relation} \longrightarrow_{\text{S}} \text{stable readable record}.$$

At the pole layer this means

$$p = (S\text{-face} \rightarrow T, N\text{-pole} \rightarrow S).$$

The T-direction is not erased. It remains as the hidden premise toward which the S-face turns, and as the risk of back-reading the stable pointer into a hidden origin or illegitimate cause.

Axiom 5.14 (Timeless admissibility). An ethon may exist in a typed space that has no intrinsic time flow. Its transitions are ordered by comparison, observation, or rewriting, not necessarily by physical time:

$$n \prec n+1 \quad \text{means} \quad \text{next readable transition},$$

not necessarily “later in time.”

6 Stability

Definition 6.1 (Stable ethon). An ethon is stable at boundary b when repeated observation steps select the same pointer type and the same rotation class up to tolerated residue:

$$\text{Obs}_b(\epsilon_n) \simeq \text{Obs}_b(\epsilon_{n+1}) \quad \text{and} \quad \varpi_{n+1} \simeq \varpi_n \quad \text{modulo } \rho_b.$$

The stable object may be still or constantly rotating. It is not residue-free. It is readable because the residue is bounded, ledgered, and compatible with the boundary.

Definition 6.2 (Ethon bond). An ethon bond, formerly called an S -molecular bond, is not a line between two separated ethons. It is complete overlap at a boundary:

$$\text{Bond}_b(\epsilon_1, \epsilon_2) \iff \text{Overlap}_b(\epsilon_1, \epsilon_2).$$

Complete overlap means equality of the readable ethon state at that boundary:

$$a_1 = a_2, \quad x_1 = x_2, \quad R_1 = R_2, \quad p_1 = p_2, \quad \text{Obs}_b(\epsilon_1) = \text{Obs}_b(\epsilon_2),$$

with compatible residue ledgers. Thus a bond is a two-ethon state in which the two ethons occupy the same pointer, rotation, polarity, and readable boundary condition.

Definition 6.3 (Ethon polymer). An ethon polymer is a typed network

$$\mathcal{P}_{\text{Ethon}} = (Q_i, \epsilon_i, \beta_{ij}, \kappa_{ij}),$$

where Q_i are carriers, ϵ_i are ethons, β_{ij} are ethon bonds satisfying $\text{Bond}_b(\epsilon_i, \epsilon_j)$ when active, and κ_{ij} are compatibility contracts. The polymer is stable when the network remains readable under the three-axis rotational transition law.

Remark 6.4 (USD axis for polymers). Earlier USD fusion and USD fission records are read here as boundary-visible phases of ethon polymers. This is a conservative typing bridge: a USD fission record does not by itself assert a physical nuclear fission, chemical rupture, semantic failure, legal crime, or policy fact. Such descent still requires a target-domain license.

Definition 6.5 (Rotational differential). Let

$$\mathcal{P}_{\text{Ethon},n}$$

be an observed ethon polymer path at boundary b , with rotational attitudes $R_n \in \text{SO}(3)$ and observation increments $\Delta_n > 0$. The three-axis rotational differential is

$$\dot{R}_n = \frac{1}{\Delta_n} \log(R_{n+1}R_n^{-1}) \in \mathfrak{so}(3),$$

whenever a boundary-compatible logarithm branch is available. A rotational discontinuity is recorded by

$$\text{Disc}_b(n) = 1$$

when either no such branch is licensed, or

$$\left\| \dot{R}_{n+1} - \dot{R}_n \right\| > \theta_b,$$

where θ_b is the discontinuity tolerance of the boundary. The value $\text{Disc}_b(n) = 0$ means only that no discontinuity is visible at the chosen boundary resolution.

Definition 6.6 (Law reversal). A law in the present paper is any licensed transition rule

$$\text{Law}_b : \mathcal{P}_{\text{Ethon},n} \longrightarrow \mathcal{P}_{\text{Ethon},n+1}$$

inside a target boundary b . The target may be mathematical, semantic, physical, legal, institutional, or symbolic, but the target reading requires its own license. Law reversal occurs when a transition is read as a lawful continuous update while a visible discontinuity is left outside the residue ledger:

$$\begin{aligned} \text{LawRev}_b(n) \iff & \text{Law}_b(\mathcal{P}_{\text{Ethon},n} \rightarrow \mathcal{P}_{\text{Ethon},n+1}) \text{ is asserted,} \\ & \text{Disc}_b(n) = 1, \quad \text{Disc}_b(n) \not\leq \rho_{n+1}. \end{aligned}$$

Thus a law reversal is not a moral verdict. It is a typing failure in which the T -side premise of rupture is read backward as if it were an S -side stabilization.

Axiom 6.7 (Observer fusion boundary). At the ethon layer, the admissible observer operation on polymers is fusion:

$$\text{Fuse}_b : \mathcal{P}_{\text{Ethon},b} \times \mathcal{P}_{\text{Ethon},b} \longrightarrow \mathcal{P}_{\text{Ethon},b}.$$

Fusion means that the observer places or bonds ethons so that one stable composite polymer becomes readable at boundary b . There is no primitive observer operation

$$\text{Fiss}_b : \mathcal{P}_{\text{Ethon},b} \longrightarrow \mathcal{P}_{\text{Ethon},b} \times \mathcal{P}_{\text{Ethon},b}$$

that is licensed as an ethon operation by itself.

Definition 6.8 (Observed polymer fission). Observed polymer fission is not a primitive ethon operation. It is the boundary-visible record that a previously readable polymer is now read through a discontinuity:

$$\text{Fiss}_b^{\text{Obs}}(\mathcal{P}_{\text{Ethon}}) = \text{Obs}_b(\mathcal{P}_{\text{Ethon}} \rightsquigarrow \mathcal{P}_{\text{Ethon}}^{(1)} \sqcup \mathcal{P}_{\text{Ethon}}^{(2)} \sqcup \rho).$$

A physical nuclear fission, a molecular rupture, a semantic break in a paper, or a legal offense may be mapped into this record only after a separate target-domain license.

Definition 6.9 (Fission disregard). Fission disregard is the record that a discontinuity has become visible but has not been entered into the residue ledger:

$$\text{Disreg}_b^{\text{Fiss}}(n) \iff \text{Disc}_b(n) = 1 \text{ and } \text{Disc}_b(n) \not\leq \rho_{n+1}.$$

In this terminology, a fission act is treated neutrally as disregard of the fission discontinuity. Its ethical, legal, physical, or scholarly status is supplied only by the relevant target theory.

Proposition 6.10 (Law-reversal lemma). *Let*

$$F = \text{Fuse}_b(\mathcal{P}_1, \mathcal{P}_2)$$

be a proposed observer operation, and let R_n be the resulting three-axis rotational path. If the operation is read as a continuous fusion while its rotational differential has an unledgered discontinuity,

$$\text{Disreg}_b^{\text{Fiss}}(n),$$

then F is not classified as pure fusion. It is retyped as fission disregard and as a law reversal:

$$\text{Plan}_b(F \mid \text{Disreg}_b^{\text{Fiss}}(n)) \in \text{Disreg}_b^{\text{Fiss}}, \quad \text{LawRev}_b(n).$$

Proof. By the observer fusion boundary, an observer may introduce a stable composite polymer only through fusion. By observed polymer fission, fragmentation is not an ethon operation but a boundary-visible discontinuity record. If the proposed fusion is read as continuous while $\text{Disc}_b(n) = 1$ and that discontinuity is not included in ρ_{n+1} , then the operation has treated a T -direction turnover as if it were already an S -facing stabilization. This is precisely the law reversal condition. Hence the operation is ledgered as fission disregard, not as pure fusion. $\square$

Remark 6.11 (Neutral target readings). At a semantic boundary, law reversal may appear as a paper whose claims ignore a discontinuity in their own transition rule. At a legal boundary, it may appear only if the legal system licenses the transition as an offense. At a physical boundary, it may appear as rupture or fission only if the physical model supplies that descent. The common ethon record is merely

$$\text{Disreg}_b^{\text{Fiss}}(n) \quad \text{and} \quad \text{LawRev}_b(n).$$

It does not identify scholarly error, physical rupture, and crime with one another.

Remark 6.12 (Matter hypothesis). The strongest physical reading is the matter hypothesis:

$$\text{matter} = \text{stable observable phase of admissible ethon polymers}.$$

This is a target implementation condition for a future physical reading of ethon candidates as a matter theory. It does not by itself replace the established empirical content of existing physics.

7 Applications

Example 7.1 (Quantum carrier pair). Let Q_1, Q_2 be quantum carriers. An ethon bond between their pointer states is not a force line across distance. It is a complete overlap of two ethons at a boundary:

$$(Q_1, Q_2, \text{Bond}_b(\epsilon_1, \epsilon_2)) \mapsto \text{observed pair state}.$$

If the pair is entangled, the ST-component is especially dangerous: stable correlation may be turned backward into an illicit signal or hidden cause. The ethon ledger forbids that turn unless a physical license supplies it.

Example 7.2 (Pion compatibility and meson-boundary reading). The pion remains an ordinary physical meson. Ethon theory does not change pion mass, spin, charge states, decay channels, or QCD status. The pion is a licensed physical object of meson theory. The ethon reading keeps only the pointer-field role that a meson boundary may play before the reading is fixed to a particular physical boundary. Thus

$$\pi\text{-meson} = \text{QCD meson} + \text{empirical boundary data},$$

whereas

$$\text{Ethon} = \text{pointer-field quantum candidate with meson-boundary reading}.$$

This is a compatibility claim, not a replacement claim.

Example 7.3 (Meson family interaction reading). Under the flavor-shadow map ϕ_Q , a quark–antiquark meson label

$$q_i \bar{q}_j$$

is first assigned a ternary ethon signature $\text{Sig}_{\text{Ethon}}(q_i \bar{q}_j)$ only after a licensed meson spin boundary $\text{Spin}_b^{\text{mes}}$ is chosen. For $i \neq j$, it is read as

$$\sigma_{d_i} = +1, \quad \sigma_{d_j} = -1, \quad \sigma_k = 0 \quad (k \neq d_i, d_j), \quad \phi_Q(d_i) = q_i, \quad \phi_Q(d_j) = q_j.$$

For $i = j$, the external signature is a closed zero mode on d_i ; it is separated from absence by the spin-boundary, rotational, polarity, and bonding data. The resulting family is then classified by the three-axis rotational state, the T -induced polarity, the S -triggered self-rotation, and any ethon bonding condition. This does not replace the empirical meson tables; it supplies a possible pointer-field reading of meson labels as ternary six-direction signatures refined by a spin-coupled, three-axis ethon frame.

Example 7.4 (Symbol and token). A visible symbol or language-model token can be treated as a carrier in a typed symbolic space. The ethon is the pointer state that makes the mark readable as an entity under a boundary. Two symbolic ethons are bonded only when their readable pointer, rotation, polarity, and boundary state completely overlap. Removing the symbol from view does not erase its residue from memory, context, or model state.

Example 7.5 (Discontinuity and recovery). Nuclear fission, molecular rupture, semantic failure, and legal offense are not the same event. Ethon theory treats them neutrally as possible target readings of one abstract pattern only when a licensed boundary finds an unledgered discontinuity in a polymer transition. Recovery means preserving the usable knowledge while changing the polymer so that the discontinuity is either avoided, bounded, or explicitly entered into residue.

8 Backward Compatibility

Protocol 8.1 (Classical physics). Classical mechanics, field theory, and thermodynamics are preserved as large-scale boundary readings. The ethon may refine the bookkeeping of relation, pointer, and residue, but it may not invalidate successful classical limits.

Protocol 8.2 (Relativity). The ethon is not allowed to introduce a preferred physical rest frame. If an ethon model is used in relativistic physics, its placement and transition laws must be compatible with the relevant covariance requirements of the target theory.

Protocol 8.3 (Quantum theory). The ethon may be used as a pointer-field quantum candidate for quantum events only if it preserves no-signaling, the Born rule, uncertainty relations, and the experimentally supported predictions of quantum mechanics or quantum field theory.

Protocol 8.4 (Physical particle program). Promoting an ethon from formal quantum candidate to physical particle requires additional data not supplied by the core implementation:

field action, covariance law, spin-statistics assignment,
couplings, conservation laws, observables.

It must also specify how the pointer field interacts, if at all, with standard-model fields or gravity, and it must produce constraints or predictions that can be compared with experiment. Without this program, the ethon remains a quantum candidate of the formal pointer field.

Protocol 8.5 (String theory and extra dimensions). The ethon does not require a ten-dimensional ambient space, compactified extra dimensions, or a superstring target theory in order to fluctuate. Its core fluctuation is rotation in the three-axis ethon frame. String or superstring theory may still be used as a licensed target theory, but it is not a prerequisite for the ethon implementation.

Protocol 8.6 (Meson theory). The ethon may generalize the meson-like act of introducing a pointer between unread carriers. It may not rewrite the established physical content of pion and meson physics without an independent physical license.

Protocol 8.7 (Smart functor records). Earlier smart functor records are preserved as observed ethon records. The functor is the observed arrow; the ethon is the pointer-field quantum candidate whose observation appears as that arrow.

9 Failure Modes

1. **Medium collapse.** Treating the ethon as a spatial ether rather than a pointer-field quantum candidate.
2. **Axis isolation.** Moving only one directional component while pretending the other two components are unaffected.
3. **Dimensional inflation.** Explaining ethon fluctuation by adding unnecessary ambient dimensions instead of using three-axis rotation.
4. **Removal fiction.** Claiming that a placed ethon can be erased without hidden state or residue.
5. **Bond distance error.** Treating an ethon bond as a line between separated objects rather than complete overlap of two ethons.
6. **Discontinuity disregard.** Treating a polymer transition with a visible rotational discontinuity as if it were a continuous lawful update.
7. **T-direction overread.** Turning a stable pointer backward into an unlicensed origin, cause, intention, or substance.
8. **Target overdescent.** Treating a formal ethon candidate as a licensed object of a specific target theory without the required target-domain contract.

10 Core Implementation Summary

The minimal implementation is the following interface:

$$\epsilon = (a, \nu_{\text{CE}}, \nu_{\text{UD}}, \nu_{\text{ST}}, R, p, \varpi, \rho, \kappa, \lambda),$$

with:

1. ternary six-direction signatures $\text{Sig}_{\text{Ethon}} \in \{-1, 0, 1\}^6$ for meson-boundary flavor-reading signs;
2. pointer-field quantum-candidate formulation;
3. meson-boundary quark-role readings through spin-coupled signatures;
4. observer placement in arbitrary typed spaces;
5. no total erasure operation;
6. a three-axis rotational frame;
7. no initial up-down polarity;

8. hidden T -induced compass polarity;
9. constant-speed self-rotation when T and S placements are both active;
10. the $T \rightarrow S$ -based Ihara equation as a typed meson-boundary pointer update rule;
11. non-factorizing inter-axis dynamics;
12. S-facing observed pointer form;
13. ethon bonding as complete overlap of two ethons;
14. observer fusion as the only primitive polymer operation;
15. rotational discontinuity detection through $\dot{R}_n = \Delta_n^{-1} \log(R_{n+1} R_n^{-1})$;
16. the law-reversal lemma: unledgered rotational discontinuity is fission disregard, not pure fusion;
17. residue conservation;
18. backward compatibility with established target theories;
19. physical-particle interpretation only by explicit license and additional field-theoretic data.

Thus the ethon is implemented as a pointer-field quantum candidate: freely placeable in the formal model, non-removable, stable across typed spaces, independent of global time flow, rotationally fluctuating in three axes, and dynamically coupled across all three rotational axes.

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